Solving Absolute Value Equations

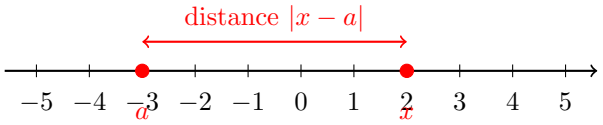
1. Introduction

The **absolute value** of a real number x, written $|x|$, is the distance from x to 0 on the number line. Because distance is never negative, $|x| \geq 0$ for every real number x.

Definition (piecewise):

$$|x| = \begin{cases} x, & \text{if } x \geq 0, \\ -x, & \text{if } x < 0. \end{cases}$$

Distance interpretation. For any real number a, $|x - a|$ is the distance between x and a on the number line.



The rule for solving $|E| = c$. Let E be any expression (like x, $x - 3$, $2x + 1$, ...) and let c be a real number.

- If $c > 0$, then $|E| = c$ means $E = c$ or $E = -c$.
- If $c = 0$, then $|E| = 0$ means $E = 0$ (one solution).
- If $c < 0$, then $|E| = c$ has **no solution** (distance can't be negative).

Three-step procedure.

1. Isolate the absolute value (get $|E|$ alone on one side).
2. Apply the definition: $E = c$ or $E = -c$.
3. Solve each linear equation. Check on a number line that solutions are at the correct distance.

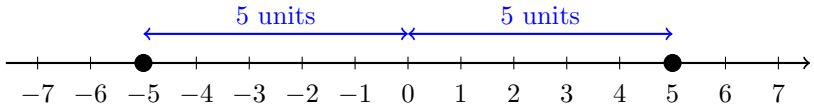
2. Two Worked Examples

Example 1. Solve $|x| = 5$.

Solution. By the definition, $|x| = 5$ means x is 5 units away from 0. So

$$x = 5 \quad \text{or} \quad x = -5.$$

Solution set: $\{-5,\ 5\}$.



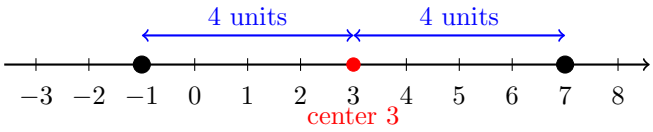
Example 2. **Solve $|x-3|=4$.**

Solution. $|x-3|=4$ says the distance from x to 3 is 4. So x is 4 units to the left of 3 or 4 units to the right of 3:

$$x-3=4 \quad \text{or} \quad x-3=-4$$

$$x=7 \quad \text{or} \quad x=-1.$$

Solution set: $\{-1,\ 7\}$.



3. Twenty Practice Problems

Solve each equation using the definition of absolute value. For each problem, draw a quick number line and mark the solutions (and the "center" a when the equation has the form $|x-a|=c$).

1. $|x|=7$
2. $|x|=0$
3. $|x|=-3$
4. $|x-2|=5$
5. $|x+4|=6$
6. $|x-1|=0$
7. $|2x|=10$
8. $|3x|=12$
9. $|2x-1|=7$
10. $|3x+2|=11$
11. $|x-5|=2$
12. $|x+7|=3$
13. $|4-x|=6$
14. $|1-2x|=5$
15. $\left|\frac{x}{2}\right|=4$
16. $|x-3|+2=9$
17. $2|x+1|=8$
18. $|3x-6|=9$
19. $|2x+5|=11$
20. $|5-2x|=7$

Answer Key

1. $x = \pm 7$
2. $x = 0$
3. No solution
4. $x = 7$ or $x = -3$
5. $x = 2$ or $x = -10$
6. $x = 1$
7. $x = \pm 5$
8. $x = \pm 4$
9. $x = 4$ or $x = -3$
10. $x = 3$ or $x = -\frac{13}{3}$
11. $x = 7$ or $x = 3$
12. $x = -4$ or $x = -10$
13. $x = -2$ or $x = 10$
14. $x = -2$ or $x = 3$
15. $x = \pm 8$
16. $x = 10$ or $x = -4$
17. $x = 3$ or $x = -5$
18. $x = 5$ or $x = -1$
19. $x = 3$ or $x = -8$
20. $x = -1$ or $x = 6$